AC209B · MILESTONE 3

Three baselines, three answers

Modeling housing mentions in SMI discharge summaries

Baselines · Inference · What we trust

PROJECT

#63 · Baseline Modeling & Inference

TEAM

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How consistently is housing mentioned in discharge summaries for patients hospitalized with serious mental illness?

WHY

Clinical guidelines

Housing should be assessed and documented at discharge for patients with serious mental illness.

WHO

SMI cohort

3,808 admissions with a primary SMI diagnosis (schizophrenia, bipolar, severe MDD).

HOW

Sentence-level classifier

Train on labeled sentences, apply to full discharge notes, aggregate to a per-note flag.

Training data: MIMIC-III labeled SDOH

WHAT IT IS

MIMIC-III-SDOH

A hand-labeled set of 1,800 clinical sentences released alongside MIMIC-III, each tagged for the social determinant of health (SDOH) it discusses — housing, food, employment, isolation, etc.

We use it as our supervised training set: it gives us ground-truth housing labels at the sentence level.

1,800

labeled sentences

80 / 20 stratified split for train / val

11.1%

of sentences mention housing

highly imbalanced — minority class

MIMIC-III

source database

publicly available, hand-labeled

Baseline 1 — Keyword search

WHAT IT IS

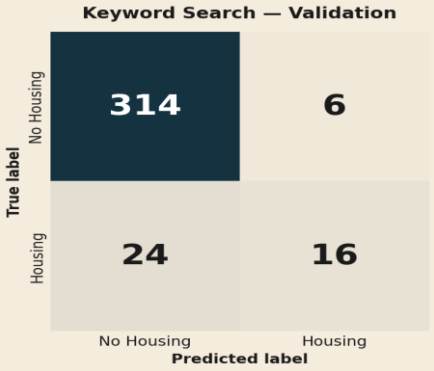
Regex match on a curated list of 23 housing-related terms.

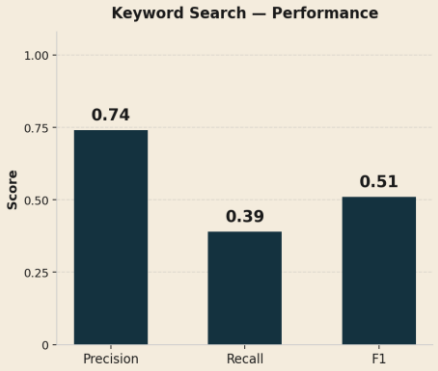
WHY A BASELINE

Simple, transparent, no training data needed. The natural floor for any ML model to beat.

KEYWORDS

homeless · shelter · street · evicted · housing instability · couch surfing · doubled up · unsheltered · …





RESULT — VALIDATION SET

Catches the obvious cases — but misses 6 in 10 real housing mentions.

A floor, not a ceiling. ML should beat this.

Baseline 2 — TF-IDF + Random Forest

WHAT IT IS

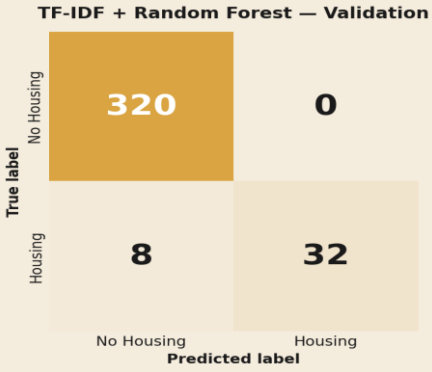
Sparse word-frequency vectors (unigrams + bigrams) fed to a tuned Random Forest classifier.

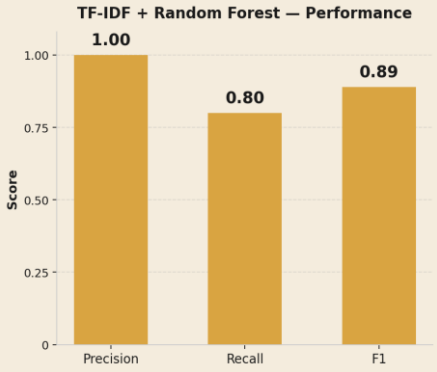
WHY A BASELINE

The simplest non-trivial way to turn text into vectors. Class-weighted to handle the 11% positive rate.

FEATURES

TF-IDF, max 5,000 features, n-grams (1,2), English stop-words removed.





RESULT — VALIDATION SET

Big jump on the validation set — recall 0.80, F1 0.89.

Beats the keyword recall floor of 0.39 by 2x.

Baseline 3 — Word2Vec + MLP

WHAT IT IS

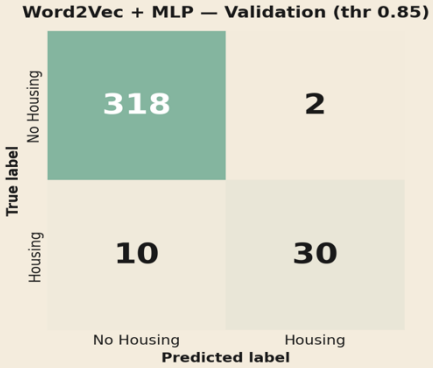
Custom Word2Vec embeddings (trained on our SMI notes), averaged per sentence, classified by an MLP.

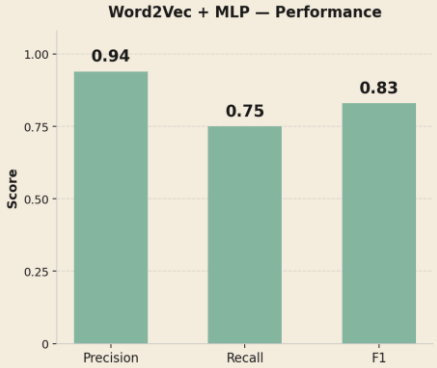
WHY A BASELINE

Moves from word counts to semantic embeddings — words like "shelter" and "motel" land near each other in vector space.

FEATURES

100-d Word2Vec → mean pool → MLP (100, 50). Threshold tuned on validation to 0.85.



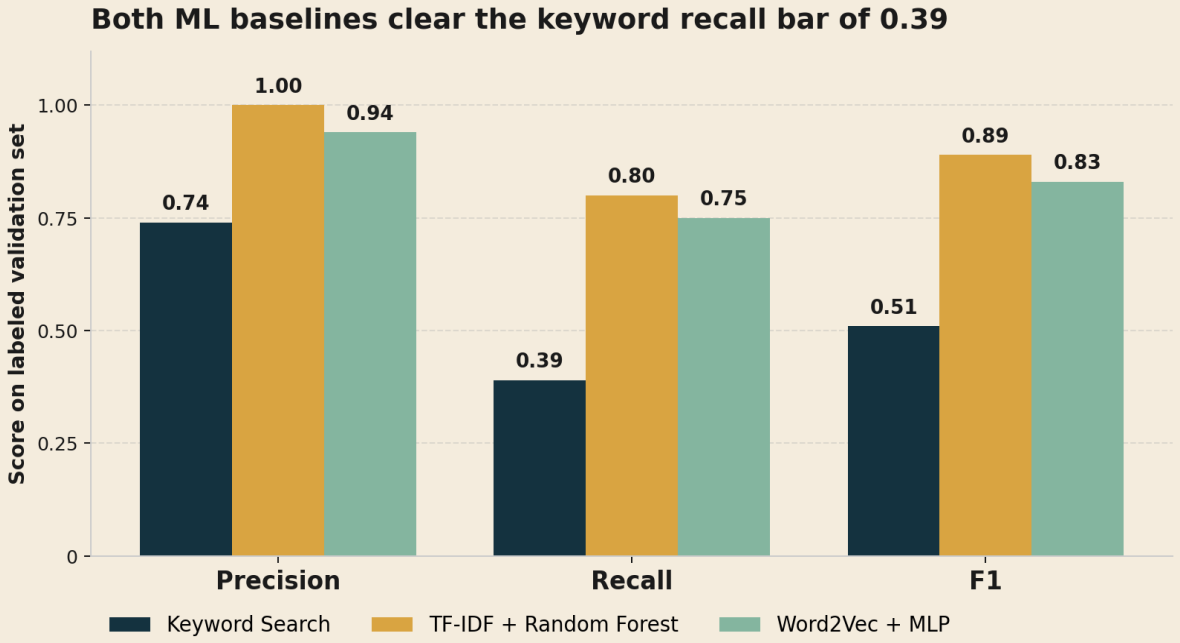


RESULT — VALIDATION SET

Strong validation performance — F1 0.83 at the tuned threshold.

Slightly lower precision than RF, but better at semantic generalization.

All three models, side by side



BEST ON VALIDATION

TF-IDF + RF

F1 = 0.89

Highest precision (1.00) and the highest F1 in the held-out validation split.

But validation performance ≠ generalization. The next slides test that.

Inference data: the MIMIC-IV SMI cohort

WHAT

MIMIC-IV

Newer, larger critical-care database (~546k admissions). Discharge notes from MIMIC-IV-Note. The cohort: admissions whose primary diagnosis is schizophrenia, bipolar disorder, or severe / recurrent MDD.

WHY

Clinical guidelines say housing should be assessed and documented at discharge for SMI patients. We want to measure how often it actually is. This is our inference data — and it is unlabeled.

3,808

admissions in our SMI cohort

PREPROCESSING

1. Split notes into sentences
2. Drop non-narrative sections (labs, meds, vitals, identifiers)
3. Keep narrative sections (HPI, Brief Hospital Course, Social Hx)
4. Aggregate sentence-level predictions to a per-note flag

Keyword search on the SMI cohort

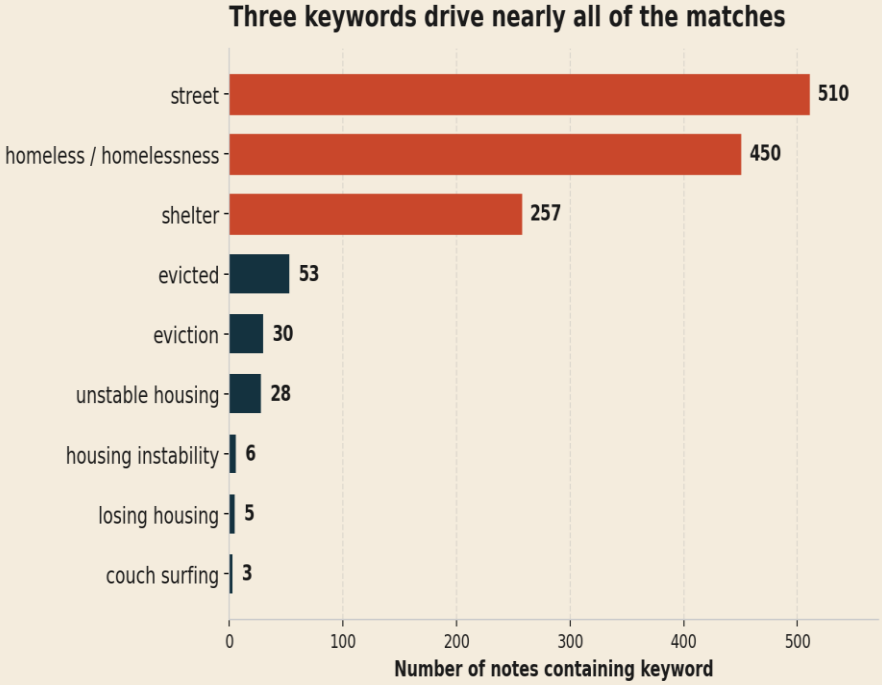
OF NOTES FLAGGED

50.9%

of SMI discharge summaries contain a regex-matched housing keyword.

THE CONCERN

Likely overcounts via false positives: "street" can mean street clothes, "shelter" can be metaphorical.



TF-IDF + Random Forest on the SMI cohort

OF NOTES FLAGGED

1.2%

of SMI discharge summaries flagged by the validation-best model.

AGREEMENT WITH KEYWORD SEARCH

Disagrees with the regex on roughly half the cohort (1,897 notes).

THE PROBLEM

Implausibly low — and disagrees with the regex on 1,897 / 3,808 notes.

WHY IT FAILS TO GENERALIZE

Bag-of-words, not semantics.

TF-IDF vectors aren't semantic — a synonym the regex never saw is a different feature, not a similar one.

Distribution mismatch.

MIMIC-III-SDOH sentences are short and synthetic; MIMIC-IV discharge sentences are messy, abbreviated, and variable in length.

Word2Vec + MLP on the SMI cohort

OF NOTES FLAGGED

83.2%

at the F1-tuned threshold of 0.85 — far above the keyword 50.9%.

DISTRIBUTION SHIFT

MIMIC-III training sentences look very different from MIMIC-IV discharge sentences — model is operating off-distribution.

ASSESSMENT

Generalizes better than RF — but probably over-flags.

WHY

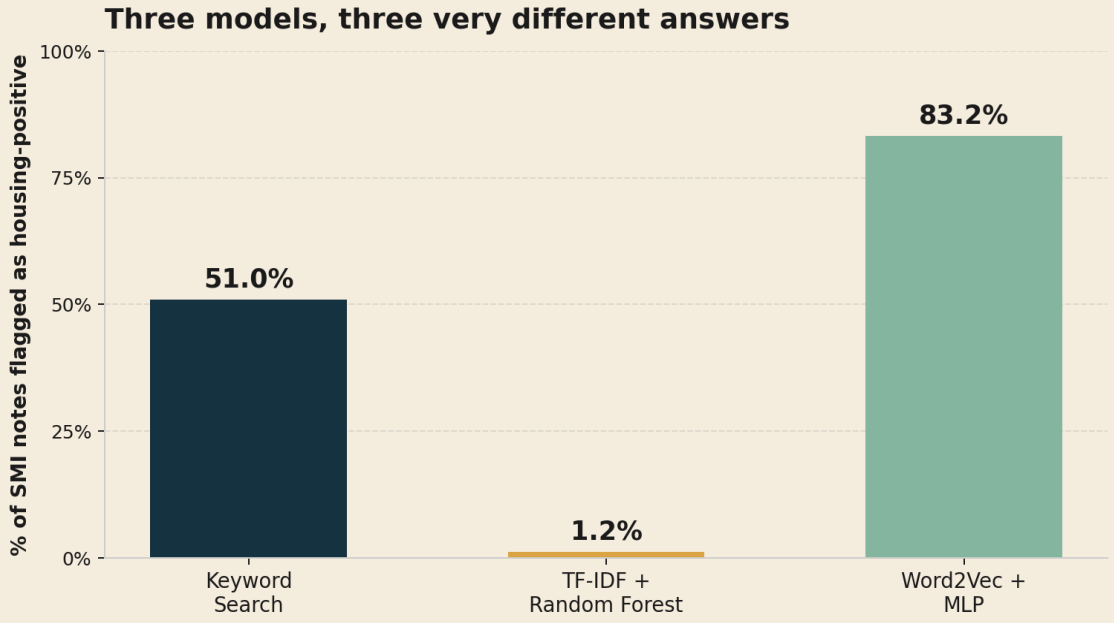
Better representation.

Semantic embeddings let the model see synonyms the regex misses.

But not state-of-the-art.

Sentence vector = mean of word vectors, so "the" weighs as much as "homeless." Motivates ClinicalBERT, where attention learns which tokens matter.

Three models, three different answers



THE PUZZLE

Same 3,808 notes. Very different answers.

RF barely flags anything (1.2%).

MLP flags almost everything (83.2%).

Both are out-of-distribution. Neither is fully trustworthy on this cohort.

Main takeaways

01 Keyword search is the most trustworthy baseline.

Recall is low, but its 50.9% rate is the most defensible estimate we have today.

02 TF-IDF + RF doesn't generalize from MIMIC-III to MIMIC-IV.

Strong on validation (F1 0.89), but flags only 1.2% of the cohort — bag-of-words can't bridge the vocabulary gap.

03 Word2Vec + MLP over-flags despite a tuned threshold.

Even at 0.85 (vs default 0.5), the model fires on 83.2% of notes. Mean-pooled embeddings + class imbalance + distribution shift.

04 The training imbalance compounds the problem.

11% positive class. Oversampling and class-weighted loss help on validation, but don't fix off-distribution behavior.

05 These limits motivate ClinicalBERT next.

Token-level attention + pretrained clinical text — the right tool for sentences this messy.

Where Milestone 4 goes

01

Fine-tune ClinicalBERT

Class-weighted loss, F1-tuned threshold.

02

Re-run subgroup analysis

Mirror the MS2 breakdowns (race, insurance, gender, diagnosis).

03

Compare ClinicalBERT vs baseline models

Compare models on the same notes

THANK YOU

Questions?

Matlin Gilman · Brianna Grissom | Project #63 · AC209B · Milestone 3